Module Schedule

<u>Key Date</u> | <u>Module 1</u>

Key Date	Module 1
1/20 Tuesday	Form groups
1/27 Tuesday	Module 1 Lab workday in class, build different microstructures.
1/29 Thursday	**Module 1 worksheet due at start of class**

Group Assignments:

#	Material Science Groups - 12 PM section				Task
1	Kurt Mokros	Carter Hire	Adrian Pigg	Nate Blasiole	1
2	Donald Dunn	David Feucht	Trevor Steinke	Jackson Rohrs	2
3	Grayson Dowell	Eli Thompson	Ben Misa	Lucas Winkelman	3
4	Lane Rellinger	Matthew Heefner	Luke Metzger	Carter Sherrick	4
5	Kaelynn Lloyd	Carter Berghauer	Justin Zhou	Gabby Burgess	5
6	Kyle Hume	Luke Stouffer	Christian Walker	Timothy Geiger	1
7	Hayden McDowell	Michael Togliatti	Trever Huber	Davis Wushinske	2
8	Taya Bucholz	Dudley Bravard	Caleb McCann	Shannon Koomen	3
9	Clayton Perry	Ethan Berei	Aaron Dowdy	Randy Blatnik	4

<table>
<tr><th>#</th><th colspan="4">Material Science Groups - 1:30 PM section</th><th>Task</th></tr>
<tr><td>1</td><td>Adam Luersman</td><td>Richard Hamm</td><td>Trent Davis</td><td>Bobby Stanyard</td><td>5</td></tr>
<tr><td>2</td><td>Connor Apple</td><td>Caden Fritz</td><td>Brandon Wise</td><td>Bryce Kopriva</td><td>4</td></tr>
<tr><td rowspan="2">3</td><td rowspan="2">Cassidy Rhoades</td><td>Matthew Wilt</td><td>Jackson Kallay</td><td rowspan="2">Ryan Harmon</td><td rowspan="2">3</td></tr>
<tr><td colspan="2">Dakota Walkowiak</td></tr>
<tr><td>4</td><td>Carter Hohenbrink</td><td>Blake Sommers</td><td>Lucas Culotta</td><td>Braedon McMillen</td><td>2</td></tr>
<tr><td>5</td><td>Adam Lawson</td><td>Brady Giesige</td><td>Connor Grolnic</td><td>Tuff Woodruff</td><td>1</td></tr>
<tr><td rowspan="2">6</td><td rowspan="2">Peyton Causby</td><td>Vincent Mertz</td><td>Marshall Blanton</td><td rowspan="2">Parker Schnippel</td><td rowspan="2">5</td></tr>
<tr><td colspan="2">Brett Bagatti</td></tr>
<tr><td>7</td><td>Joshua Talamas</td><td>Matt McKeown</td><td>Michael Brown</td><td>Garrett Buchanan</td><td>4</td></tr>
<tr><td>8</td><td>Mylie Kifus</td><td>Kaden Montross</td><td>Nathan Roasa</td><td>Cooper Coughlin</td><td>2</td></tr>
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Module 1: Atomic structure of engineering materials

Each group will do one task from the followings:

Task 1. Body Centered Cubic (BCC) and Face Centered Cubic (FCC) unit cells

Task: Build BCC and FCC unit cells. Illustrate

- how many atoms (foam balls) in each unit cell,
- coordination number in each unit cell,
- the relationship between side length and atomic radius.

List common examples of BCC and FCC metals on a sheet a paper next to your unit cells..

Please watch the following video and complete your pre-lab assignment before the lab workday.

- BCC, FCC (2:38): https://www.youtube.com/watch?v=uwf8FHlvWKA
- Concept definition (2:33): https://www.youtube.com/watch?v=oIevxVKgJFg&t=16s
- BCC (3:30): https://www.youtube.com/watch?v=_9RnbGqtkd4
- FCC (3:32): https://www.youtube.com/watch?v=GSPVC34ijIA

2. Hexagonal Closed Packed (HCP) unit cell

Task: Build an HCP unit cell. Illustrate

- how many atoms (foam balls) in each unit cell,
- coordination number in each unit cell,
- the relationship between side length and atomic radius.

List common examples of HCP metals on sheet of paper next to your HCP Cell

Please watch the following video and complete your pre-lab assignment before the lab workday.

- HCP (4:01): https://www.youtube.com/watch?v=_OhQv5gsoSQ
- Concept definition (2:33): https://www.youtube.com/watch?v=oIevxVKgJFg&t=16s

3. Relative sizes of ions in the crystal structure of ceramics

Build linear, triangular, tetrahedral, and octahedral structure and discuss why the crystal structures of ceramics are more complicated than metals. Use a caliper to measure the size of foam balls to make sure the ratio of cation and anion is in the range in the table below. In addition, build NaCl unit cell as shown.

On a sheet of paper, write how cations and anions arrange in each structure. Give common examples of ceramics.

Please watch the following videos and complete your pre-lab assignment before the lab workday.

- Ceramic crystal structure (6:06): https://www.youtube.com/watch?v=BAiRs1IYgMs

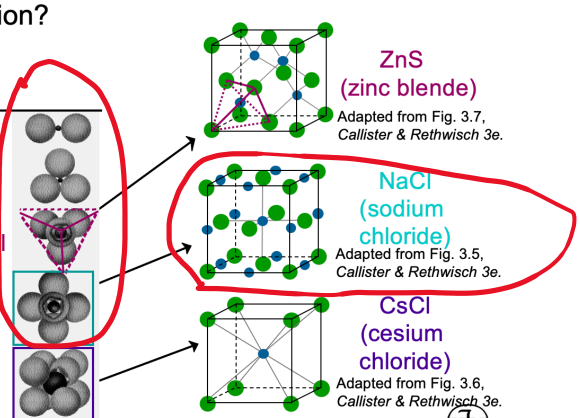
Ceramics - Coordination # and Ionic Radii

- Coordination # increases with $\frac{r_{cation}}{r_{anion}}$

To form a stable structure, how many anions can surround around a cation?

$\frac{r_{cation}}{r_{anion}}$	Coord #	
< 0.155	2	linear
0.155 - 0.225	3	triangular
0.225 - 0.414	4	tetrahedral
0.414 - 0.732	6	octahedral
0.732 - 1.0	8	cubic

Adapted from Table 3.3, *Callister & Rethwisch 3e.*



Adapted from Fig. 3.7, *Callister & Rethwisch 3e.*

Adapted from Fig. 3.5, *Callister & Rethwisch 3e.*

Adapted from Fig. 3.6, *Callister & Rethwisch 3e.*

Chapter 3 - 13

4. Graphene and Graphite

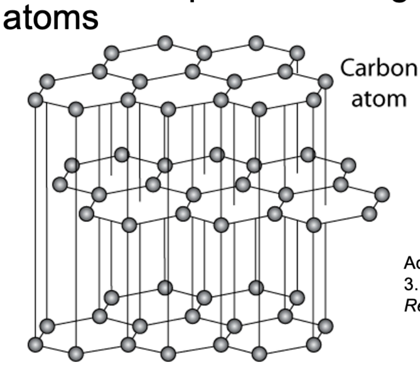
Build layered structure of graphene and graphite. Illustrate why planes slide easily over one another in graphite. Discuss why graphite is a good solid lubricant.

Please watch the following videos and complete your pre-lab assignment before building the structures.

Graphene (3:53): https://www.youtube.com/watch?v=frgetR7qJec

Graphite

– layered structure – parallel hexagonal arrays of carbon atoms



Adapted from Fig. 3.17, *Callister & Rethwisch 3e.*

– weak van der Waal's forces between layers

– planes slide easily over one another -- good lubricant

5. Body-Centered Tetragonal unit cell (BCT)

Build a BCT unit cell and illustrate how many atoms in the unit cell.

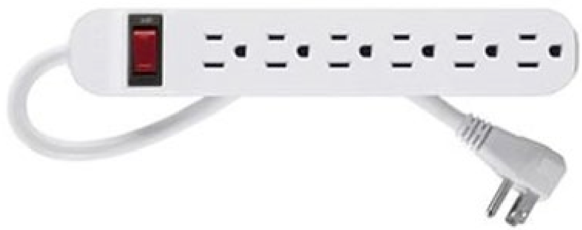
On a sheet of paper, describe the side length of the unit cell. Discuss the mechanical properties of the BCT metal (Martensite).

Please read following websites and complete your pre-lab assignment before building the structures.

- https://msestudent.com/body-centered-tetragonal-unit-cell/
- https://www.sciencedirect.com/topics/engineering/body-centered-tetragonal

Lab equipment:

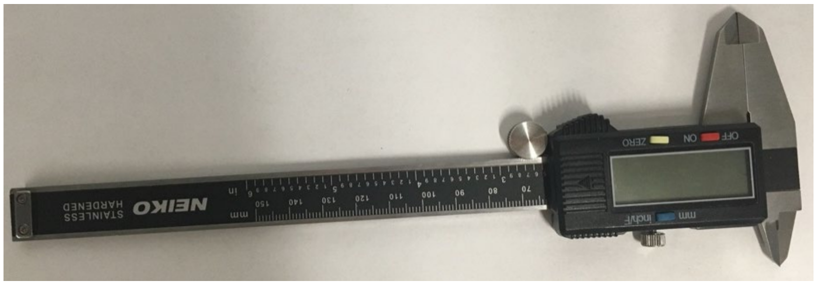
1. Power strip: plug in the hot wire foam cutters and hot glue guns



2. Hot wire foam cutter: cut the foam balls

3. Hot glue gun: glue foam balls together

4. Caliper: measure the diameter of foam balls for Task 4, ceramic crystal structures

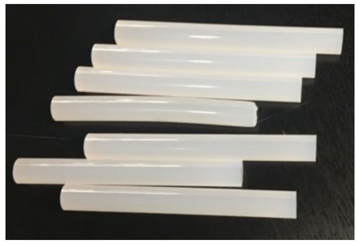


Lab material supplies:

1. Foam balls with various sizes

2. Craft sticks: to form the covalent bonds in graphene and graphite structure

3. Glue sticks for hot glue guns



4. Strings: can be used to form Van der Waals bonds between layers in graphite structure

5. Rubber bands: can be used to form Van der Waals bonds between layers in graphite structure

6. Cardboard to cover the lab tables to protect the surface of tables